



Department of Education

STEM Education
National Schools of Excellence
STEM Chemistry Syllabus
Grade 11
Term 2

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

Published in 2021 by the Department of Education, Papua New Guinea

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ISBN

Acknowledgements

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

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SECRETARY'S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

THEMA	MODULE	TOPICS
BIODIVERSITY	Ethnobotany	1. Medicinal/Herbal Plants 2. Bioactive compounds in plants 3. Examples of Therapeutic Drugs from plants
BIOTECHNOLOGY	Principles of Biotechnology	1. Polymerase Chain Reaction 2. Dyes in DNA Sequencing 3. Proteins
HEALTH AND MEDICINE	Biochemistry in Medicine	1. Drug Design 2. Drug Pharmacology 3. Significance of Diagnostic Laboratory Tests
EARTH SCIENCE		
WASTE MANAGEMENT		
INFRASTRUCTURE		
ENERGY		
BATTERY TECHNOLOGY		
INFORMATION SYSTEM		
COMPUTER SCIENCE		
ARTIFICIAL INTELLIGENCE		
ENTERPRENEURSHIP		

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM Domain

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – TERM 2

Table 1.1: Thema Sequence and Duration

Grade 11 Term 2: STEM-proper Course		
BIODIVERSITY (3 Weeks)	11.1 Ethnobotany	1. Medicinal/Herbal Plants 2. Bioactive Compounds in Plants 3. Examples of Therapeutic Drugs from nature
BIOTECHNOLOGY (3 Weeks)	11.1 Principles of Biotechnology	1. Polymerase Chain Reaction 2. Dyes in DNA Sequencing 3. Proteins
HEALTH AND MEDICINE (3 Weeks)	11.1 Biochemistry in Medicine	1. Drug Design 2. Drug Pharmacology 3. Significance of Diagnostic Laboratory Tests

Table 1.2 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Biodiversity	Broad Content Standard	
Module 11.1	Ethnobotany	11.1a: Scientifically document medicinal plants in Papua New Guinea 11.1b: Discuss bioactive compounds and biomolecules from plants 11.1c: Identify and discuss therapeutic drugs from nature	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Medicinal/Herbal Plants	11.1.1: Scientifically document medicinal plants in Papua New Guinea	Students' ability to demonstrate knowledge and skills in; - Medicinal plants documentation in PNG - Important bioactive compounds and biomolecules from plants - Therapeutic drugs derived from nature	Assignment + Written Test
2. Bioactive Compounds in Plants	11.1.2a: Define bioactive compounds 11.1.2b: Describe the uses of bioactive compounds in plants 11.1.2c: Discuss different techniques for separation of biomolecules from plant extract 11.1.2d: Test crude plant extracts against a target 11.1.2e: Observe their effects against the target		
3. Examples of Therapeutic Drugs from Nature	11.1.3a: Identify therapeutic drugs from nature 11.1.3b: Discuss the importance of the therapeutic drugs.		

Thema	Biotechnology	Broad Content Standard	
Module 11.1	Principles of Biotechnology	11.1 Discuss Principles of Biotechnology in Polymerase Chain Reaction, Dyes DNA Sequencing and Proteins in medical, agricultural and industrial environments	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Polymerase Chain Reaction	11.1.1: Describe the components, procedures and applications of polymerase chain reaction	Students' ability to demonstrate knowledge and skills in; • Components, procedures and applications of polymerase chain reaction in health and medicine • Dyes In DNA sequencing • Proteins as important macromolecule in living functions.	Assignment + Written Test
2. Dyes in DAN sequencing	11.1.2a: Discuss the importance of dye chemistry in gene sequencing 11.1.2b: Discuss different sequencing procedures		
3. Proteins	11.1.3a: Define proteins 11.1.3b: Identify types of proteins 11.1.3c: Discuss synthesis of proteins 11.1.3d: Discuss case studies of proteins utilized in biotechnology		
Thema	Health and Medicine	Broad Content Standard	
Module 11.1	Biochemistry in Medicine	11.1a: Discuss types of drug design, 11.1b: Expound on how drugs work 11.1c: Signify diagnostic laboratory results	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Drug Design	11.1.1: Discuss types of drug design	Students' ability to demonstrate knowledge and skills in; • Types of drug design • How drugs work • Significance of Diagnostic laboratory results	Assignment + Written Test
2. Drug Pharmacology	11.1.2: Describe clearly how drugs work in terms of mechanism of action, absorption, distribution, metabolism and elimination and potential toxic effects		
3. Significance of Diagnostic laboratory tests	11.1.3a: Discuss the types of diagnostic laboratory tests. 11.1.3b: Describe the significance of diagnostic laboratory tests.		

Table 1.3 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	1	1	1
Biodiversity/ 11.1 Ethnobotany	11.1a: Scientifically document medicinal plants in Papua New Guinea 11.1b: Discuss bioactive compounds and biomolecules from plants 11.1c: Identify and discuss therapeutic drugs from nature	✓		
Biotechnology/ 11.1 Principles of Biotechnology	11.1 Discuss Principles of Biotechnology in Polymerase Chain Reaction, Dyes in DNA Sequencing and Proteins in medical, agricultural and industrial environments		✓	
Health and Medicine/ 11.1 Biochemistry in Medicine	11.1a: Discuss types of drug design, 11.1b: Expound on how drugs work 11.1c: Signify diagnostic laboratory results			✓

THEMA

BIODIVERSITY

INTRODUCTION

Biodiversity is short for biological diversity. It includes all organisms, species, and populations; the genetic variation among these; and all their complex assemblages of communities and ecosystems. It also refers to the interrelatedness of genes, species, and ecosystems and their interactions with the environment. Biodiversity is a key measure of the health of any ecosystem, and of our entire planet. Every organism in an ecosystem, or biome, relies on other organisms and the physical environment. For example, plant and animal species need each other for food, and depend on the environment for water and shelter. Biodiversity describes how much variety an ecosystem has, in terms of resources and species and also genetically within species. A more diverse ecosystem will have more resources to help it recover from famine, drought, disease or even the extinction of a species. There are several levels of biodiversity, each indicating how diverse the genes, species and resources are in a region.

In this Chemistry Phylum, the focus is on ethnobotany in an effort to introduce students to the knowledge of the traditional uses of botany (plants) for different purposes but with specific emphasis on medicinal uses or herbal products.

MODULE 11.1: ETHNOBOTANY

Native people all over the world have traditional uses of plants for various purposes, such as for medicines, foods, intoxicants, clothing, building, and so on. The scientific study of the relationships between native people and the traditional uses of the plants is known as Ethnobotany. Ethnobotanists are the scientists that undertake these studies and document the existing relationships for proper scientific purposes.

In this module, the students will be required to list plants that are of use to their traditional settings and properly document them for further scientific studies. The emphasis is on the traditional uses of plants for medical purposes.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11.2: Types of Chemical Reactions

Broad Content Standard

11.1a: Scientifically document medicinal plants in Papua New Guinea

11.1b: Discuss bioactive compounds and biomolecules from plants

11.1c: Identify and discuss therapeutic drugs from nature

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- Medicinal plants documentation in PNG
- Important bioactive compounds and biomolecules from plants
- Therapeutic drugs derived from nature

Topic 1: Medicinal/ Herbal Plants

Performance Indicators;

Students are expected to:

11.1.1: Scientifically document medicinal plants in Papua New Guinea

Content Expansion

Body of Knowledge for their understanding

- Documentation of medicinal plants in Papua New Guinea
 - Title
 - Background
 - Methods
 - Results
 - Discussions
 - Conclusion

Set of skills for their proficiency levels

- Identify and Scientifically document medicinal plants in Papua New Guinea
- Discuss results
- Present results in class
- Submit results for possible assessment and data base records

Trait of Attitudes and Values skills to govern their lives

- Appreciate native medicinal plants in Papua New Guinea.
- Adopt and value the uses of native medicinal plants in Papua

Topic 2: Bioactive Compounds in Plants

Performance Indicators;

Students are expected to:

11.1.2a: Define bioactive compounds

11.1.2b: Describe the uses of bioactive compounds in plants

11.1.2c: Discuss different techniques for separation of biomolecules from plant extracts

11.1.2d: Test crude plant extracts against a target

11.1.2e: Observe their effects against the target

Content Expansion

Body of Knowledge for their understanding

- Bioactive compounds - phytochemicals, which can be extracted from plants or plants by-products, and able to regulate metabolic functions leading to beneficial effects
- Bioactive compounds in plants.
 - primary metabolites
 - proteins
 - fats
 - sugars
 - Secondary metabolites
 - Alkaloid
 - Flavonoids
 - Phenolic acid
 - Terpenes
 - Saponins
 - Tannins
- Pharmacological properties of common plant bioactive compounds

Compound type	Pharmacological properties
Terpenes	Antimicrobial, antiviral, antihypertensive, anthelmintic, antibacterial, anticancer, antimalarial, anti-inflammatory
Phenolics acids	Ant-carcinogenic and anti-mutagenic, anti-inflammation and anti-allergic
Alkaloids	Antispasmodic, antimalarial, analgesic, diuretic activities, local anesthetic, antihypertensive, antiasthma, anti-malarials, diuretic, bactericidal
Flavonoids	Antioxidant activity, cardiovascular protective, anti-inflammatory, hepatoprotective, antiviral, antibacterial
Saponins	Antitumor, antiviral, antifungal, anti-inflammatory, immunostimulant, anti-hypoglycemic, anti-hepatotoxic and hepatoprotective, anticoagulant, neuroprotective, antioxidant
Tannins	Antioxidant, anti-carcinogenic, diuretics, hemostatic, anti-mutagenic, antiseptic,

- Characterization of Bioactive Molecules
 - Extract plant materials from Violaceae or Fabaceae (Legume) plants
 - Test these extracts against the targets – mosquito larvae or small fish
 - Observe and record their effects on the targets
 - Undertake a dose response trial and plot the results on a graph

Set of skills for their proficiency levels

- Describe the bioactive compounds in plants
- Discuss pharmacological aspects of bioactive compounds
- Identify and discuss the effects of the extracts on the targets
- Present discussions

Trait of Attitudes, Values and skills to govern their lives

- Appreciate the uses of plants in Papua New Guinea
- Acknowledge the importance of PNG's traditional plants in medicine
- Justify pharmacological aspects of traditional plants (bioactive compounds)
- Appreciate that traditional plants have significant uses in terms of disease treatment.
- Appreciate some plants can be used to manufacture modern pharmaceutical drugs used in hospitals

Topic 3: Examples of Therapeutic Drugs from Nature

Performance Indicators;

Students are expected to:

11.1.3a: Identify therapeutic drugs from nature

11.1.3b: Discuss the importance of the therapeutic drugs.

Content Expansion

Body of Knowledge for their understanding

Table 1: Significant Plant-Derived Pharmaceutical Products		
Source Plant	Drug	Indication
Salix spp	Aspirin	Analgesic Anti-pyretic cardiovascular
Atropa Belladonna	Astropine	Anti-Cholinergic Pupil Dilation
Digitalis Lanata	Digoxin	Cardiotonic
Mucuna Deeringiana	(L)-dopa	Anti-parkinsonism
Dioscorea Deltoidea	Diosgenin	Anti-fertility
Papaver Somniferum	Morphine Codein	Analgesic and Antitussive
Colchicum Autumnale	Colchicine	Anti-tumour Anti-gout
Cinchona Ledgeriana	Quinine	Anit-Malarial
Catharantus Roseus	Vincristine Vinblastine	Anti-tumour

Table 2
Some more recent new products and drugs in development

SOURCE PLANT	PLANT PRODUCT*	DERIVATIVE*	INDICATIONS	MODE OF ACTION
<i>Galanthus woronowii</i>	galanthamine (Shire/Johnson & Johnson)		Alzheimer's	reversible cholinesterase inhibitor
<i>Podophyllum peltatum</i>	podophyllotoxin	etoposide (Bristol Myers Squibb/Novartis)	anti-cancer	topoisomerase II inhibitor
<i>Camptotheca acuminata</i>	camptothecin	topotecan (SmithKline Beecham) irinotecan (Pharmacia/ Aventis)	anti-viral anti-cancer	topoisomerase I inhibitor
<i>Artemisia annua</i>	artemisinin	artemether (Aventis)	malaria	schizontocide
<i>Taxus brevifolia</i>	taxol (Bristol Myers Squibb)	docetaxel (Aventis)	anti-cancer	microtubule disaggregation inhibitor
<i>Tricosanthes kirilowii</i>	tricosanthin		anti-cancer	reverse transcriptase inhibitor
<i>Coleus forskolii</i>		forskolin	anti-viral glaucoma impotence	adenylate cyclase stimulator
<i>Cannabis sativa</i>	Δ^9 -tetrahydrocannabinol (Unimed) cannabidiol	nabilone (Eli Lilly) Δ^9 -tetrahydrocannabinol hemisuccinate	antiemetic in chemotherapy	analgesic sedative muscle relaxant
<i>Silybum marianum</i>	silymarin (Madaus)		appetite stimulation multiple sclerosis spasticity pain management hepatic disorders amanita poisoning	free radical scavenger

* = company affiliation denotes marketed product. Source: various, see additional source material

- Quinine
 - Plant origin – cinchona ledgeriana
 - Indication/uses – anti-malarial
 - Case study of discovery (refer to Teacher's Resource Book)
- Artemether (Aventis)
 - Plant origin – Artemisia annua
 - Indication/uses – antimalarial
 - Case study of discovery (refer to Teacher's Resource Book)

Set of skills for their proficiency levels

- Identify the plant origin of therapeutic drugs
- Discuss the uses of therapeutic drugs
- Present Discussions

Trait of Attitudes, Values and skills to govern their lives

- Appreciate the plant origin of therapeutic drugs in nature
- Acknowledge the use of therapeutic drugs from nature
- Appreciate the importance therapeutic drugs

Application of Knowledge & Skills (Career Focus)

- Pharmaceutical industries
- Biochemists
- Doctors
- Pharmacists
- Ethnobotanists
- Laboratory scientists

THEMA

BIOTECHNOLOGY

INTRODUCTION

Biotechnology combines biology with technology to engineer, modify or upgrade the parts of biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry and engineering to modify the biological systems. There are four (4) major areas biotechnology is applied – medical, agricultural, industrial and environmental.

MODULE 11.1: PRINCIPLES OF BIOTECHNOLOGY

There are two important techniques which enable development of modern biotechnology. This module will introduce students to these techniques. The first technique is the polymerase chain reaction (PCR) which is the main aspect of chemical engineering and is discussed in Topic 1. Dye chemistry in gene sequencing (chemical engineering) is discussed in Topic 2. Protein expressions which are biological molecules will be covered in Topic 3.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11.2: Types of chemical reactions
- Module 11.5: Chemical Bonding

Broad Content Standard

11.1: Discuss Principles of Biotechnology in Polymerase Chain Reaction, Dyes in DNA Sequencing and Proteins in medical, agricultural and industrial environments

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- Components, procedures and applications of polymerase chain reaction in health and medicine
- Dyes in DNA sequencing
- Proteins as important macromolecule in living functions.

Topic 1: Polymerase Chain Reaction

Performance Indicators;

Students are expected to:

11.1.1: Describe the components, procedures and applications of polymerase chain reaction (PCR).

Content Expansion

Body of Knowledge for their understanding

- Define Polymerase Chain Reaction (PCR)
 - History of polymerase chain reaction (PCR)
- Components of PCR
 - DNA template
 - DNA polymerase
 - Primers
- Procedures
 - Denaturation
 - Annealing
 - Extension
- Applications
 - Forensic cases
 - Anthropology
 - Evolution
 - Medical diagnosis
 - Infectious Disease (e.g. SARS-Cov2-2019 or COVID-19)
 - Research

Set of skills for their proficiency levels

- Describe the components, procedures and applications of polymerase chain reaction
- Discuss the importance of the applications of PCR

Trait of Attitudes, Values and skills to govern their lives

- Appreciate the significance of polymerase chain reaction
- Value polymerase chain reaction as a diagnostic test for diseases
- Plan appropriate treatment from PCR results

Topic 2: Dyes in DNA Sequencing

Performance Indicators;

Students are expected to:

11.1.2a: Discuss the importance of dye chemistry in gene sequencing

11.1.2b: Discuss different sequencing procedures

Content Expansion

Body of Knowledge for their understanding

- Define dye chemistry
- Define;
 - DNA
 - DNA sequencing
 - Gene
 - Gene sequencing
 - Genome
 - Genome sequencing
 - Genomics
- Different sequencing procedures
 - Wandering spot analysis
 - Chemical cleavage method
 - Chain termination method

Set of skills for their proficiency levels

- Study purpose of Dye Chemistry
- Discuss different procedures of sequencing
- Present discussions

Trait of Attitudes, Values and skills to govern their lives

- Appreciate that Polymerase Chain Reaction can be used to solve forensic crimes
- Acknowledge that plants can be modified genetically for cultivation or commercial purposes

Topic 3: Proteins

Performance Indicators;

Students are expected to:

11.1.3a: Define proteins

11.1.3b: Identify types of proteins

11.1.3c: Discuss synthesis of proteins

11.1.3d: Discuss case studies of proteins utilized in biotechnology

Content Expansion

Body of Knowledge for their understanding

- Define proteins
- Different types of proteins
 - Glycoproteins
 - Cry proteins
 - Protein functions;
 - Antibodies (Immunoglobulin G)
 - Enzymes (Phenylalanine hydroxylase)
 - Messenger (Growth hormone)
 - Structural components (Actin)
 - Transport/ storage (Ferritin)
- Synthesis of proteins
 - Transcription and translation.
- Case study of Proteins utilized in Biotechnology
 - Cry proteins for insect control
 - GAT protein for herbicide tolerance

Set of skills for their proficiency levels

- Define proteins
- Discuss protein types and functions
- Outline protein synthesis
- Study case studies of protein utilization and do presentations

Trait of Attitudes and Values skills to govern their lives

- Appreciate proteins as important macromolecule to sustain life.
- Acknowledge numerous functions of proteins as important to keep living things alive.

Application of Knowledge & Skills (Career Focus)

- Doctors
- Laboratory scientists
- Genetic engineers
- Pharmaceutical industries
- Biochemists
- Biomedical engineers

THEMA HEALTH AND MEDICINE

INTRODUCTION

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

Furthermore, the recent emergence of SARS-Cov2–2019 (COVID-19) pandemic has shut down the world and caused severe economic downturn globally; it has disrupted the global supply chain. It has exposed the deficiency within the Papua New Guinean health delivery services and their political decision-making processes. This is not the first global outbreak of novel viral diseases. There have been similar outbreaks in the recent past, such as Middle-East Respiratory Syndrome (MERS), Severe Acute Respiratory Syndrome (SARS), Avian Influenza (H1N1 and its variants), African Swine Flu, and so on. It is therefore of paramount importance to educate and prepare the next generation of medical doctors and scientists with sound knowledge in medicine to combat emergence of future pandemics.

MODULE 11.1: BIOCHEMISTRY IN MEDICINE

Chemistry has many relations to human life. It provides really fundamental knowledge of health and disease. Our bodies are composed of chemical substances and all the numerous functions of the living organisms depend, in one way or another, on chemical processes. Chemical processes enable us to digest food, keep warm and supply us with energy. Even the impressions of our senses, thoughts of our brains, as well as the mode of conveying these through the nerves, are more or less dependent on chemical processes.

Chemistry has made major contributions to health care. The development of new drugs involves chemical analysis and synthesis of new compounds. This is evident in the many recent advertisements made on the large number of new drugs produced.

This module calls attention very briefly to some of the ways in which chemistry may be able to help the theory and practice of medicine.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11.2: Types of chemical reactions
- Module 11.3: Masses, moles and molarity
- Module 11.4: Acids, bases and salts
- Module 11.5: Chemical Bonding

Broad Content Standard

11.1a: Discuss types of drug design,

11.1b: Expound on how drugs work

11.1c: Signify diagnostic laboratory results

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- Types of drug design
- How drugs work
- Diagnostic laboratory results

Topic 1: Drug Design

Performance Indicators;

Students are expected to:

11.1.1: discuss types of drug design

Content Expansion

Body of Knowledge for their understanding

- Types
- Ligand-based design
 - Receptors
 - Biological activity of molecules

Structure-based design

- Target sites
- Binding affinity and selectivity

Set of skills for their proficiency levels

- Discuss types of drug design

Trait of Attitudes and Values skills to govern their lives

- Appreciate that therapeutic drugs can be designed to treat diseases to improve human lives

Topic 2: Drug Pharmacology

Performance Indicators;

Students are expected to;

11.1.2: Describe clearly how drugs work in terms of mechanism of action, absorption, distribution, metabolism and elimination and potential toxic effects

Content Expansion

Body of Knowledge for their understanding

- Mechanism of action
 - Drug (ligand) – a chemical substance which produces a biological effect when administered to a living organism.
 - Receptor - specialized target macromolecule that binds a drug (ligand)
 - Types of receptor;
 - enzymes,
 - nucleic acids, or
 - specialized membrane-bound proteins
 - drug-receptor complex
- Absorption
 - Sites of absorption
- Distribution
 - How drugs are distributed
- Metabolism
 - Site of metabolism
- Elimination
 - Routes of elimination
- Potential toxic or adverse effects
 - Common adverse effects

Set of skills for their proficiency levels

- Describe clearly the mechanism of action and the absorption, distribution, metabolism and the elimination and potential side effects of drug.

Trait of Attitudes and Values skills to govern their lives

- Appreciate how drugs work so appropriate drugs are administered for the appropriate ailments such that drugs improve health and lives of humans.

Topic 3: Significance of Diagnostic Laboratory Tests

Performance Indicators;

Students are expected to;

11.1.3a: Discuss the types of diagnostic laboratory tests.

11.1.3b: Describe the significance of diagnostic laboratory tests.

Content Expansion

Body of Knowledge for their understanding

- Full Blood Count
 - White blood cells
 - Neutrophils
 - Eosinophils
 - Basophils
 - Monocytes
 - Red blood cells
 - Transport of oxygen
 - Structure of hemoglobin
 - Oxygen binding site on Hemoglobin
 - Factors affecting oxygen binding affinity
 - Platelets
 - blood clots
- Liver function test – enzyme activities
 - alanine transaminase (ALT)
 - aspartate aminotransferase (AST)
 - alkaline phosphatase (ALP)
 - albumin
 - bilirubin
- Kidney function test
 - Urinalysis (can be done in the chemistry laboratory using multi/ dip sticks).
 - Parameters (multi/ dip sticks)
 - Acidity
 - Concentration
 - Proteins
 - Sugar
 - Ketones
 - Bilirubin
 - Evidences of infection – nitrites/leukocyte esterase
 - Blood
 - interpretation and significance of the results

Set of skills for their proficiency levels

- Describe the types of diagnostic laboratory tests.
- Interpret the results of the tests
- Explain the significance of diagnostic laboratory tests.

Trait of Attitudes, values and skills to govern their lives

Appreciate;

- The types of diagnostic laboratory tests
- The interpretation and significance of laboratory tests
- How to execute a specific diagnostic test to confirm an ailment to improve the management of patients.

Application of Knowledge & Skills (Career Focus)

- Doctors
- Pharmacists
- Nurses
- Pharmaceutical industries
- Biochemists
- Medical laboratory scientists

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